

4. Halogenoalkanes are hydrolysed by aqueous sodium hydroxide. State and explain which of 1-fluoropropane, 1-chloropropane and 1-bromopropane is hydrolysed most rapidly. [2]

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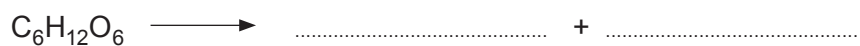
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5. (a) State the type of polymerisation involved when 2-fluorobut-2-ene,  $\text{CH}_3\text{CF}=\text{CHCH}_3$ , forms a polymer. [1]

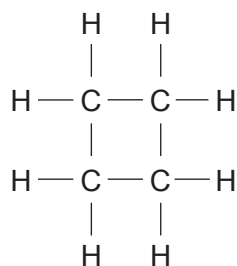
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- (b) Draw **one** repeat unit of the polymer formed when 2-fluorobut-2-ene is polymerised. [1]

6. Complete the equation for the formation of ethanol by the fermentation of glucose. [1]



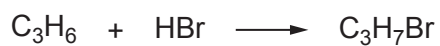
4. Cyclobutane is a cyclic hydrocarbon whose structure is shown.



Draw the structures of **two** structural isomers of this compound.

[2]

5. Propene reacts with hydrogen bromide.



Draw the structures of the **two** isomers formed and give the name of the major product.

[2]

Major product .....

